SIMATS ENGINEERING
ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME:	Cloud Computing and Big Data Analytics	**COURSE CODE:**	CSA1516

COURSE OUTCOME COVERED
CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2) **Total Marks: 50**
Question Count: 5

Code of Conduct
I Y Sravan Kumar (192472289) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: ___SravanKumar_________

Assessment Tasks
Analytical Problem Solving
1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical application. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded:

1. Compare Type-1 and Type-2 Hypervisors

Type-1 Hypervisor

- Runs directly on physical hardware.
- Does not require a host operating system.
- Provides better performance because there is less overhead.
- Provides stronger security because it has a smaller attack surface.
- Suitable for centralized management of large numbers of VMs.
- Examples: VMware ESXi, Microsoft Hyper-V, Xen.

Type-2 Hypervisor

- Runs on top of a host operating system.
- Has additional overhead because requests pass through the host OS.
- Security depends partly on the security of the host OS.

- Easier to install and use on personal computers.
- Examples: VirtualBox, VMware Workstation.

Better choice

For a cloud datacenter hosting mission-critical applications, Type-1 hypervisor is better because it provides higher performance, stronger isolation/security, reliability, and better enterprise manageability.

2. Workload Isolation Using Virtualization

A cloud provider can run finance, healthcare, and student portal workloads on the same physical server by placing them in separate virtual machines (VMs).

How virtualization provides isolation

- Each VM has its own virtual CPU, memory, storage, and network interfaces.
- The hypervisor separates the VMs from one another.
- A problem or failure in one VM should not directly affect another VM.
- Virtual networking can further separate traffic using virtual switches, VLANs, and security policies.

For example:

```
     Physical Server
          |
       Hypervisor
    /     |      \
 Finance   Healthcare Student
   VM        VM        VM
```

Two major attack surfaces

1. Hypervisor attacks: An attacker may exploit a hypervisor vulnerability to escape from a VM or access other VMs.
2. Virtual network attacks: Improper network configuration can expose communication between isolated workloads.

Therefore, hypervisor patching, access control, network segmentation, and monitoring are important.

3. Analyze the VM Host Utilization

Given:

- CPU = 92%
- Memory = 88%
- Storage I/O latency = 35 ms
- Average VM boot time = 170 seconds

Analysis

CPU – 92%:
CPU utilization is very high, indicating that the host may not have enough processing capacity for all running VMs.

Memory – 88%:
Memory utilization is also high. This can cause memory pressure and swapping, which reduces VM performance.

Storage latency – 35 ms:
High storage I/O latency indicates a possible storage bottleneck. Slow disk operations can affect applications and VM startup.

VM boot time – 170 seconds:
The long boot time can result from high CPU/memory utilization and slow storage.

Corrective actions

- Migrate some VMs to less-loaded hosts.

- Add additional CPU and RAM resources.
- Reduce VM over-provisioning.
- Upgrade slow storage to faster SSD/NVMe storage.
- Use load balancing and resource monitoring.
- Investigate VMs generating excessive storage I/O.

Likely bottlenecks: CPU, memory, and particularly storage I/O are under pressure, contributing to poor VM performance and long boot times.

4. Software Defined Datacenter vs Traditional Datacenter

A Software Defined Datacenter (SDDC) virtualizes and manages computing, storage, and networking resources through software.

1. Compute Virtualization

- Physical servers are divided into multiple VMs.
- VMs can be quickly created, resized, migrated, or deleted.
- Computing resources can be allocated dynamically.

2. Storage Virtualization

- Storage from multiple physical devices can be combined into logical storage pools.
- Storage capacity can be dynamically allocated according to application requirements.
- Reduces dependence on individual storage hardware.

3. Network Virtualization

- Network functions are controlled using software.
- Virtual networks can be created without manually changing physical network infrastructure.
- Enables faster network configuration and automated security policies.

Agility advantage

Traditional datacenters require more manual hardware configuration, whereas SDDCs use software, virtualization, and automation to provision and modify resources quickly.

Therefore, SDDC provides greater agility, scalability, automation, and resource utilization.

5. Multi-Cloud Federation Identity Problem

Problem

In a multi-cloud environment, different cloud providers may use different:

- User identities
- Authentication mechanisms
- Access-control policies
- Identity formats

This can cause identity mismatches, preventing users from securely accessing resources across different cloud providers.

Proposed secure design

1. Federated Identity Management:
Use a common Identity Provider (IdP) so that users can authenticate once and access authorized services across participating cloud providers.

2. Standard protocols:
Use standards such as SAML, OAuth 2.0, and OpenID Connect for secure identity and authentication exchange.

3. Single Sign-On (SSO):
Users authenticate once and can access authorized resources across multiple clouds without maintaining separate passwords for every provider.

4. Role-Based Access Control (RBAC):
Assign permissions according to user roles and apply the least-privilege principle.

5. Multi-Factor Authentication (MFA):
Require an additional authentication factor for sensitive cloud resources.

6. Privacy protection:
Share only the minimum identity information required between providers and encrypt identity information during transmission and storage.

Conclusion

A combination of federated identity, SSO, standard authentication protocols, MFA, RBAC, encryption, and data minimization can resolve identity mismatches while maintaining security and user privacy across multiple cloud providers.